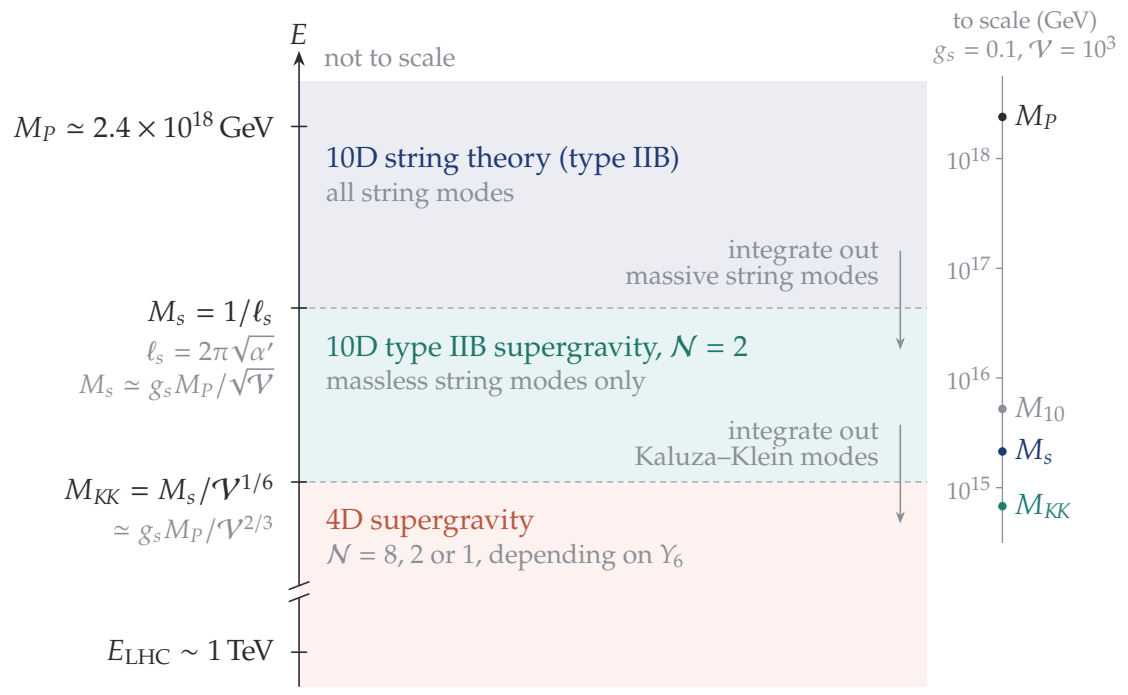

E
not to scale
to scale (GeV)
$g_s = 0.1$, $\mathcal{V} = 10^3$
$M_P \simeq 2.4 \times 10^{18}$ GeV
10D string theory (type IIB)
all string modes
integrate out
massive string modes
$M_s = 1/\ell_s$
$\ell_s = 2\pi\sqrt{\alpha'}$
$M_s \simeq g_s M_P/\sqrt{\mathcal{V}}$
10D type IIB supergravity, $\mathcal{N} = 2$
massless string modes only
integrate out
Kaluza–Klein modes
$M_{KK} = M_s/\mathcal{V}^{1/6}$
$\simeq g_s M_P/\mathcal{V}^{2/3}$
4D supergravity
$\mathcal{N} = 8$, 2 or 1, depending on Y_6
$E_{\rm LHC} \sim 1$ TeV
M_P
10^{18}
10^{17}
10^{16}
M_{10}
M_s
10^{15}
M_{KK}